



Galactic Explorers: Unsupervised Learning

A cosmic adventure through clustering, guided by the galaxy's most curious crew





The Great Cosmic Mystery

Our space raccoons have just arrived at a brand-new galaxy — and it's utterly uncharted.



Millions of Objects

Stars, planets, and nebulae everywhere — but none of them have names or labels.



No Teacher Here

Nobody told the raccoons what's what. They must figure it out themselves.



The Big Question

How do we make sense of infinite, unlabelled chaos?



CORE CONCEPT

What is Unsupervised Learning?

It's the art of finding structure **without a teacher**. There are no labels like "cat" or "dog" — just raw, unlabelled data.

- The goal is to discover hidden patterns or "neighborhoods" in the data — like finding clusters of similar stars in a galaxy.

Meet the Clustering Logic

Clustering is the raccoons' primary navigation tool. It groups similar space objects together based on shared traits — no labels required!



Group Similar Objects

Stars that share properties naturally fall into the same cluster.



Measure Similarity

We use **Euclidean distance** to decide how close two stars really are.



No Labels Needed

The algorithm discovers the groups all by itself — pure exploration!

The Workhorse: K-Means Algorithm

The raccoons pick **K random spots** in the galaxy to set up temporary base camps (centroids), then refine them iteratively.

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Step 1: Assign

Each star checks which base camp is closest and joins that group.

2

Step 2: Move

The base camps shift to the center of their local group.

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Step 3: Repeat

Keep going until the raccoons stop moving — convergence!



PRO TIPS

Refining the Hunt

Randomly picking base camps can lead to poor groupings. Two tricks help our raccoons do better:

K-Means++

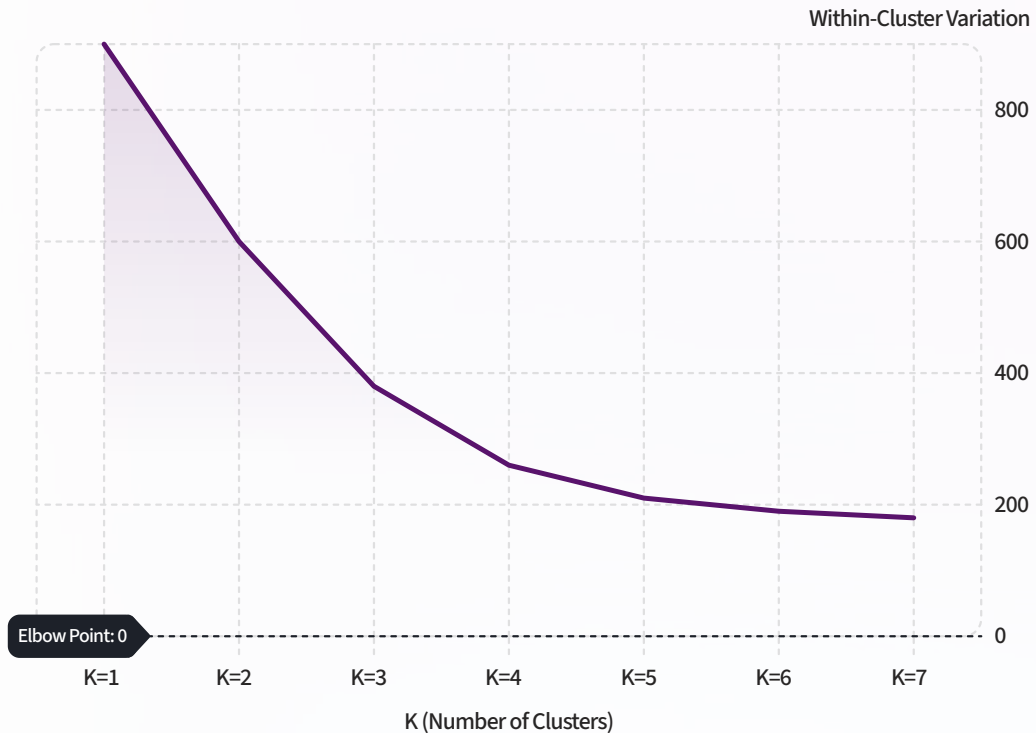
Smarter initial starting points — camps are chosen to be far apart, avoiding bad clustering.

Standardize Your Data

Always scale your features so different units don't confuse the raccoons' distance calculations.



The Elbow Method



How many base camps should the raccoons set up? The **Elbow Method** plots within-cluster variation against K and looks for the "sweet spot" — where adding another cluster gives **diminishing returns**.



The "elbow" is where the curve bends — that's your optimal K! 🦨

Beyond K-Means: Hierarchical Clustering

Sometimes the raccoons want to see the **big picture** — not just fixed groups, but how everything relates.

Build a Tree

A **dendrogram** shows how stars merge into clusters at different levels of similarity.

Cut Where You Like

Choose your desired number of clusters by "cutting" the tree at the right height.

No K Required

Perfect when we don't know how many clusters we need in advance.





ADVANCED TOOL

Specialised Tools: DBSCAN

What about stray asteroids or nebulae that don't fit anywhere? **DBSCAN** is the raccoons' solution for messy, real-world data.

→ Density-Based

Identifies high-density regions as clusters, ignoring sparse noise.

→ Any Shape

Finds natural clusters of **any shape** — not just neat spheres like K-Means.

→ Outlier Detection

Labels lonely objects as noise — perfect for spotting anomalies! 🛸

Mission Accomplished! 🚀

From chaos to clarity — our space raccoons have successfully mapped the entire galaxy using the power of unsupervised learning.

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Core Algorithms

K-Means, Hierarchical, and DBSCAN
explored

∞

Patterns Found

Hidden structures discovered in
unlabelled data

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Big Takeaway

Raw data becomes actionable insight
through clustering

✓ Ready to start your own data exploration? The universe is waiting! 🐾

